

Team Details

TEAMNAME: TEAM GALAXY

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PROBLEM STATEMENT

Improper waste segregation is a major problem in urban areas, leading to inefficient recycling processes. Develop a system that encourages households and communities to segregate waste and track recycling efforts.

1. Waste classification system
2. Household participation tracking
3. Recycling rewards or gamification system
4. Data analytics dashboard for municipalities
5. Awareness and reporting platform

Technologies Used

Frontend

- Next.js – React framework used to build the web interface
- React – Component-based UI development
- Tailwind CSS – Styling and responsive design
- TypeScript – Type-safe JavaScript for better code reliability

Backend / Database

- Firebase – Backend platform used for authentication and database
- Cloud Firestore – Real-time database to store waste reports and user data

Blockchain

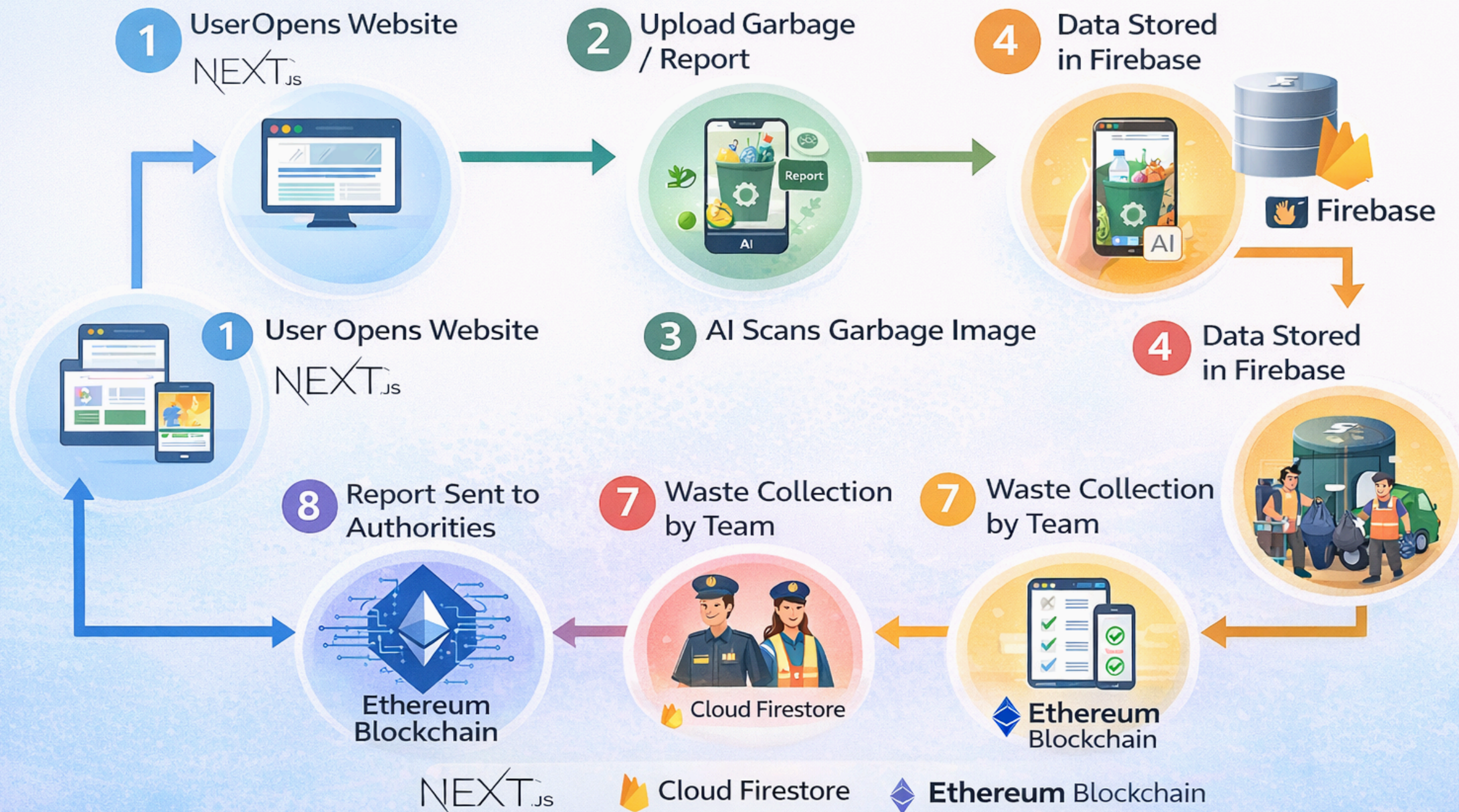
- Ethereum – Blockchain network used to store waste tracking records
- Smart Contracts – Used to create secure and transparent transactions
- Web3.js – Connects the website to blockchain

AI / Image Processing

- Computer Vision Model – Detects garbage from images
- Machine Learning Model – Classifies waste types (plastic, organic, metal, etc.)

Workflow

Workflow of the Smart Waste System



Thank You